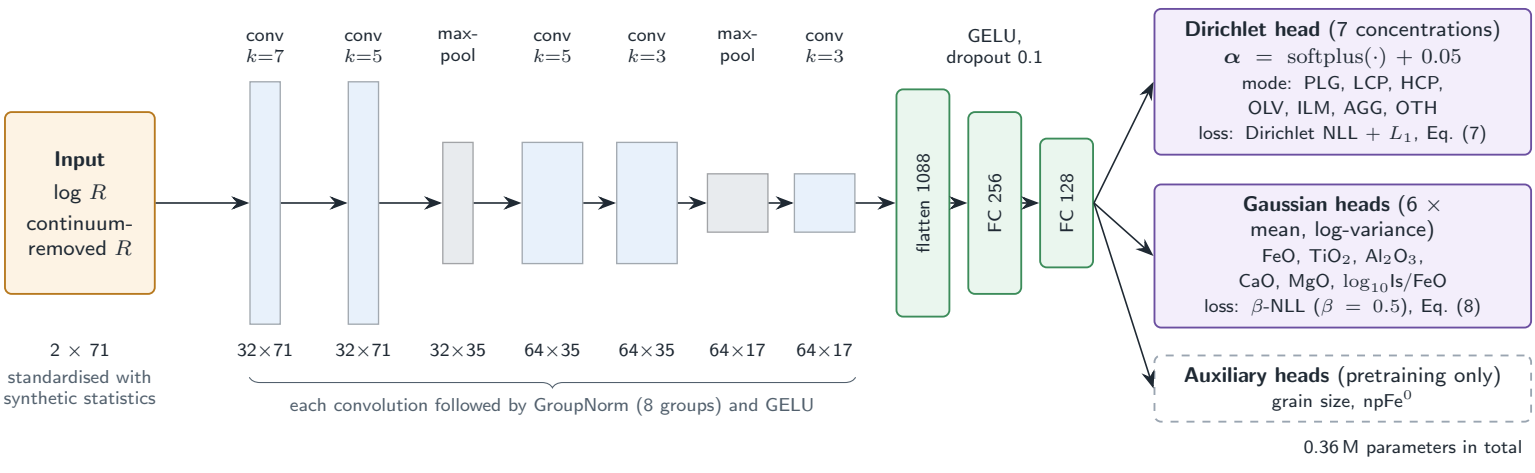

Input
log R
continuum-removed R
2 × 71
standardised with synthetic statistics
conv k=7
conv k=5
max-pool
conv k=5
conv k=3
max-pool
conv k=3
32×71
32×71
32×35
64×35
64×35
64×17
64×17
each convolution followed by GroupNorm (8 groups) and GELU
GELU, dropout 0.1
flatten 1088
FC 256
FC 128
Dirichlet head (7 concentrations)
$\boldsymbol{\alpha} = \mathrm{softplus}(\cdot) + 0.05$
mode: PLG, LCP, HCP, OLV, ILM, AGG, OTH
loss: Dirichlet NLL + L_1, Eq. (7)
Gaussian heads (6 × mean, log-variance)
FeO, TiO_2, Al_2O_3, CaO, MgO, $\log_{10}$Is/FeO
loss: β-NLL ($\beta = 0.5$), Eq. (8)
Auxiliary heads (pretraining only)
grain size, npFe^0
0.36 M parameters in total